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# =====
# Lab 3: Cholera Outbreak Case Tracking
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# 1. Create Dataset
days   <- c("Mon", "Tue", "Wed", "Thu", "Fri")
cases  <- c(50, 65, 80, 40, 90)
status <- c("Low", "Medium", "High", "Low", "High")

status   <- factor(status, levels = c("Low", "Medium", "High"))
case_data <- data.frame(days, cases, status)
print(case_data)

# =====
# 2. Conditional Alerts
# =====
for (i in 1:length(cases)) {
  if (cases[i] > 70) {
    print(paste(days[i], "- High Risk"))
  } else {
    print(paste(days[i], "- Normal"))
  }
}

# =====
# 3. Calculate Growth Rate
# =====
growth <- c()

for (i in 2:length(cases)) {
  rate  <- (cases[i] - cases[i-1]) / cases[i-1]
  growth <- c(growth, rate)
}

print(growth)

# =====
# 4. Visualization
# =====

# Line graph
plot(cases, type = "l", col = "red", lwd = 2,
     main = "Cholera Outbreak Cases",
     xlab = "Days", ylab = "Cases")

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# Bar chart
barplot(cases, names.arg = days,
        col = "orange", main = "Daily Cases")
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